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The Ethics of Collecting Data via Autonomous Vehicles

Introduction/Background

Autonomous vehicles collect data using various sensors. These sensors include cameras that record passengers and the vehicles' surroundings. However, some people are concerned this leads to privacy invasion.

Autonomous vehicles need to collect data to function correctly. If autonomous vehicles are not allowed to collect any data, autonomous vehicles are not allowed to exist. However, their passengers want to be asked to consent to their data being collected and used. This issue is important because it affects the future development of autonomous vehicles.

There is a history of privacy concerns over data collection by technology. The movement for data privacy goes as far back as the 1970s (Claravine, n.d.). Privacy concerns over autonomous vehicles specifically is an extension of this larger movement. Autonomous vehicles recently became relevant in this discussion such as when the autonomous vehicle company Waymo disclosed collected data to the California Public Utilities Commission in 2019 (Vosen, 2019). This is an early example of the issue around autonomous vehicle data collection.

Stakeholder Analysis

The two major stakeholders for this issue are Waymo and the communities that Waymo operates in. Waymo is a company that deploys autonomous vehicles in cities and runs a rideshare service with those vehicles. This issue also applies to other autonomous vehicle companies, but the focus is on Waymo for simplicity. The communities that Waymo operates in refers to the

people that live in the cities where Waymo offers services with autonomous vehicles. These people are stakeholders even if they have never ridden in a Waymo autonomous vehicle.

Stakeholder 1

Values. Waymo's priorities mostly revolve around gaining trust. The company claims its "mission is to be the world's most trusted driver" (Waymo, n.d.). Therefore, its priority is to gain the public's trust. Waymo (n.d.) also claims it "[aims] to offer freedom of movement for all, create a sustainable, efficient transportation ecosystem, and make the planet better than [the company] found it." The company states it also values accessibility and transit. However, its statement regarding trust is larger than the previous quote on its website (Waymo, n.d.). Therefore, Waymo likely prioritizes gaining trust above its other values.

Position. Waymo's position on the issue is that it already takes data privacy seriously. Waymo states its "mission is to be the world's most trusted driver, and that means . . . fostering an environment where riders and communities understand and trust how [the company handles] their information" (The Waymo Team, 2026). The company claims it values customer privacy regarding their data. Waymo cares about the facts because it wants people to trust it and its products. If the company demonstrates it does not care about the facts, their customers would notice and lose trust in it.

Claims. Waymo makes claims to support the idea that it works to protect people's privacy. The company claims it "[collects] what [data it needs] to operate safely and to improve [the Waymo] service" (The Waymo Team, 2026). Waymo implies it collects the minimum amount of data required for its business. This is a fact type of claim because the company states its claims regarding how much data it collects are true. The Waymo Team (2026) claims "[it strives] to continually develop and improve the tools that help [its] riders access, manage and,

where appropriate, delete their data.” The company’s evidence that backs up its fact claim is that it shows riders the data it collects. This way, riders can see how much data Waymo collects.

Stakeholder 2

Values. The communities that Waymo operates in prioritize privacy and consent for data to be collected and transparency for what Waymo does with that data. Alessandro Acquisti (as cited in Neuman, 2026) claims "it is not clear the extent to which passengers [of autonomous vehicles] . . . are reminded that when they step into the car, . . . they are not told in its entirety how the data [from passengers] will be used." This highlights the concern that those who ride in autonomous vehicles are unaware of what happens to the data the vehicles collect from them.

Position. These communities want to be asked for consent before autonomous vehicles collect their data. They argue that if they are asked for consent, it implies they understand the risks associated with their data being collected and accept them if they provide consent. They care about the facts and events because their data being shared could lead them to face negative consequences. These risks, which will be discussed shortly, highlight how consent can reduce harm.

Claims. These stakeholders claim the data that autonomous vehicles collect from them can be used in ways that hurt them. One example is when autonomous vehicle passengers were arrested based on footage taken inside the vehicle (Neuman, 2026). This is a fact-type claim because it insists that this incident happened. One may argue that the passengers may not have been arrested had they understood the legal consequences of their actions in the autonomous vehicle being recorded and seen by law enforcement.

Argument Question

Should autonomous vehicles be required to ask for consent from communities to collect their data and disclose how companies use that data?

Stakeholder Argument

Stakeholder 1

Position on Argument. Waymo has addressed privacy concerns regarding its data collection and usage. It aims to convince the public that the company takes such matters seriously. According to Waymo, autonomous vehicles are already doing enough to protect data privacy and therefore do not need to be regulated further in this regard.

Ethical Framework. This stakeholder uses the ethical egoism framework. Elizabeth Mohn (2020) defines this framework as "a theory that claims people should make choices that are beneficial to their own self-interest." In other words, the framework emphasizes making decisions based on how the outcome benefits oneself. Mohn (2020) attributes Adam Smith and Bernard Mandeville for helping to shape ethical egoism. Mohn (2020) explains "some supporters of ethical egoism claim that if all people in a society care for themselves and work toward gaining things for themselves, society would run smoothly and people would not need to depend on others." This explains that ethical egoism aims to promote independence.

Argument. Waymo's position falls into the ethical egoism framework even when the company points out the benefits of its services. Waymo (n.d.) claims it "is committed to holding safety to a higher standard. Because when [it does], [the company makes] space for people to get around in a whole new way." While this statement explains the benefits from Waymo's services, this does not stop the company from having an ethical egoist perspective. Ethical egoists may act in ways that help other people as long as those actions also serve themselves (Mohn, 2020). While Waymo points out how its actions help communities, they still use an ethical egoism

framework because it acts that way to gain trust. This benefits the company because it gets closer to completing its mission and therefore follows the ethical egoism framework.

Course of Action. According to Waymo's perspective, no further action is necessary to protect its riders' data privacy. The Waymo Team (2026) explains the company's data policies, how those policies protect people's privacy, and how the company complies with existing privacy laws. This implies the company already takes data privacy seriously and therefore does not need to be regulated further in that regard. Waymo has likely made this information public to gain further trust from people, thus upholding the ethical egoism framework.

Stake. If autonomous vehicle data collection is further regulated to ensure consent from those whose data is collected, Waymo could lose trust from those people. A new law that requires Waymo to uphold higher privacy standards would imply its current standards are inadequate and need government intervention for improvement. This means passengers would think Waymo does not care enough about privacy to improve those standards on its own, so the company loses trust from those passengers. As mentioned earlier, the company's "mission is to be the world's most trusted driver" (Waymo, n.d.). Therefore, Waymo losing the public's trust matters to it because less trust goes against its mission.

Stakeholder 2

Position on Argument. The communities that Waymo operates in have expressed concern over the company's practices. They are worried about what its data collection implies for them. According to the communities that Waymo operates in, autonomous vehicles should be required to ask for consent to collect data and disclose how that data is used regardless of Waymo's current practices.

Ethical Framework. This stakeholder uses the utilitarianism framework. The Santa Clara Markkula Center for Applied Ethics (2014) attributes Jeremy Bentham as the framework's developer. The same source defines this framework as "a moral principle that holds that the morally right course of action in any situation is the one that produces the greatest balance of benefits over harms for everyone affected" (The Santa Clara Markkula Center for Applied Ethics, 2014). This implies everyone's overall happiness is a major principle of the utilitarianism framework.

Argument. This stakeholder's position is utilitarian because they are unhappy about the cameras on Waymo vehicles and the footage those cameras take. A Nashville resident named Rachel (as cited in Taylor, 2026) expresses "I don't like that at all. [A Waymo autonomous vehicle] is parked outside of my house all the time . . . It's just another of these private companies spying on [people]." This resident is unhappy over a Waymo vehicle's presence and perceived surveillance. Sentiments such as these relate to the utilitarian framework because they show people being less happy and thus going against utilitarianism.

Course of Action. Making autonomous vehicles ask for consent to collect data would increase happiness in those who feel such vehicles have monitored them without their consent. As stated earlier, this stakeholder includes entire communities and not just autonomous vehicle riders. In other words, people who live in homes that Waymo vehicles park near are part of the stakeholder group. People who are upset over this phenomenon want to stop it. If they have the option to decline consent for autonomous vehicles to park near their homes, they may be happier.

Stake. If autonomous vehicles do not need consent to collect data from the communities they operate in, those communities could lose the safe feeling of knowing the vehicles they encounter are not recording them and using the recordings in harmful ways. Rachel (as cited in

Taylor, 2026) describes this situation using the word “spying.” This invokes a negative emotion and implies she feels unsafe around autonomous vehicles. Acquisti (as cited in Neuman, 2026) claims "The privacy problems arise when and if driverless carrier companies [have used privacy to report and protect laws] as a pretext for blanket, indiscriminate accumulation of identifiable data for unspecified future purposes." This highlights the concern for transparency in how data collected by autonomous vehicles is used. The statement shows how unknown data usage can negatively impact the people involved in the data.

Student Position (1 page)

Opinion. Both stakeholders have something to lose from this issue. However, if a decision on this issue is made in Waymo’s favor, more harm will be done. Autonomous vehicles should be required to ask for consent to collect data from communities and disclose how the companies that operate such vehicles use the data.

Alignment. This position aligns with the second stakeholders because it acknowledges that autonomous vehicles collecting data from people without their consent and using it in ways that are not disclosed to people is harmful. As discussed, the communities that face this issue feel unsafe around autonomous vehicles because the communities have not consented to being recorded and are unaware of how the recordings are used. This goes against the utilitarian framework that this stakeholder uses. If Waymo is required to ask for consent to collect data from people, it would reduce this harm.

Recommendation. As a recommendation for how to solve this issue, there should be federal policies that hold Waymo’s leaders accountable for data collected without consent and failure to disclose how the company uses its collected data. This will affect the issue because the

threat of criminal charges will motivate Waymo's leaders to continuously improve the company's data and privacy practices. Scott Neuman (2026) has interviewed someone who "compares monitoring a robotaxi via cameras to a human taxi driver keeping an eye on passengers in the rearview mirror." However, this comparison does not account for how some autonomous vehicle passengers are unaware that they are being recorded. Irina Raicu (as cited in Neuman, 2026) claims "With all these recording devices, [people do not] see them, [and they are] not these obvious things being stuck in our faces." This shows how riding in an autonomous vehicle with cameras one is unaware of is not the same as riding in a car when one knows the human driver is watching them. There have already been examples of federal actions preventing tech companies from harming the public such as how "The Federal Trade Commission [has sent] refunds . . . to consumers as the result of a settlement with Ring [an Amazon subsidiary] over charges the company allowed employees and contractors to access consumers' private videos" (Federal Trade Commission, 2024). This is an example of how federal action can reduce harm done by tech companies. Therefore, federal policy that would hold Waymo accountable for collecting data without consent and not disclosing how the company uses the data would ensure residents of any United States areas that Waymo autonomous vehicles operate in can rest easy knowing that the vehicles will not infringe upon their privacy.

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